

Computational tractability of the Monoterpene Indole Alkaloid Biosynthesis in *Catharanthus roseus*: A pathway-level analysis

Alexander John Milner

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Abstract

Monoterpene indole alkaloids (MIAs) from *Catharanthus roseus* include the anticancer compounds vinblastine and vincristine, whose biosynthesis involves a complex network of specialized enzymatic reactions. This study evaluated the computational tractability of the MIA pathway using RetroPath2.0, enzyme characterization, and protein sequence analysis. A total of 28 chemical transformations were benchmarked, of which 15 of 21 evaluable reactions were recovered (71.4%), while seven produced technical errors. Twenty-nine pathway enzymes were assessed experimentally, with nine classified as well-characterized and 20 as partially characterized. Sequence analysis of 28 proteins identified DAT as the most divergent among the selected enzymes, while retaining characteristic BAHD acyltransferase motifs. Integration of these results highlighted technical difficulties in the iridoid branch and unresolved late-pathway steps, particularly vincristine formation. Overall, the analysis identifies both computational and experimental limitations that may constrain reconstruction and engineering of the *C. roseus* MIA pathway.

1 Introduction

Monoterpene indole alkaloids (MIAs) are a large and chemically diverse class of specialized plant metabolites produced primarily by species of the order Gentianales[1]. Among these compounds, two stand out particularly, vinblastine and vincristine, produced by *Catharanthus roseus*. These compounds were first thought to have possible anti-diabetic effects and were studied in that context in the late 1950s[2]. Subsequent research revealed their significant antimitotic activity, inhibiting tubulin polymerization and thereby inhibiting cell division at metaphase. Both compounds were rapidly adopted into clinical oncology and remain frontline treatments for Hodgkin’s lymphoma, acute leukemia, and several solid tumors today[3, 4].

The biosynthesis of vinblastine and vincristine in *C. roseus* is one of the most complex and specialized metabolic pathways known in plants, comprising more than 30 enzymatic steps distributed across multiple cell types and subcellular compartments [1, 5, 6]. The pathway emerges from the convergence of two very distinct branches. The indole branch producing tryptamine from tryptophan and the iridoid branch producing secologanin from geranyl diphosphate (GPP) through multiple enzymatic steps [1, 5]. Secologanin and tryptamine then converge to produce strictosidine, the universal precursor for all MIAs [1]. The biochemical pathway continues through multiple complex steps to create catharanthine and vindoline,

which are essential precursors for the bisindole pathway leading to vinblastine and vincristine [6, 7].

Despite decades of research, the biosynthetic pathway to vinblastine has only recently been largely elucidated, most notably with the complete iridoid branch characterized by Miettinen et al. in 2014 [5]. However, the final steps of the pathway producing vinblastine and vincristine remain unresolved [6, 7]. The enzymatic conversion of vinblastine to vincristine, which involves N-demethylation and formylation of the indoline nitrogen, has not been attributed to any characterized enzyme in *C. roseus* to date. Similarly, the peroxidase-mediated coupling of catharanthine and vindoline to anhydrovinblastine, while attributed to PRX1, remains incompletely characterized at the mechanistic level. Engineering the complete pathway in a heterologous microbial host therefore remains an outstanding challenge, partly because missing enzymatic steps cannot currently be reconstituted [6, 7].

This study presents a computational analysis of the MIA pathway in *Catharanthus roseus*, combining a RetroPath2.0 benchmark across 28 chemical transformations with a pathway-wide characterization of 29 enzymes and a sequence and homology analysis covering 28 proteins. The objective was to identify reactions that are computationally recoverable, assess the degree of experimental characterization of their corresponding enzymes, and integrate these results to highlight potential bottlenecks for pathway reconstruction and metabolic engineering.

2 Methods

2.1 RetroPath2.0 pathway benchmarking

The MIA pathway was decomposed into individual precursor-product transformations. A total of 28 chemical transformations were included in the final benchmark. RetroPath2.0 was run locally [8] using the parsed RetroRules reaction-rule dataset (`retrorules_parsed.csv`) [9], with a maximum pathway length of one reaction because the objective was to assess recovery of individual known biochemical transformations rather than reconstruct complete multi-step routes. Reaction rules were applied across the available diameter range in the local implementation. Each known product was evaluated against its expected precursor.

Recovery was assessed from the RetroPath output using the **In Sink** field. A reaction was classified as *Recovered* when the expected precursor was found among the generated products, *Not recovered* when RetroPath generated products but failed to recover the expected precursor, and *Technical error* when the workflow failed during rule generation or pathway construction.

For multi-substrate reactions, such as the condensation of secologanin and tryptamine or the coupling of catharanthine and vindoline, recovery was assessed according to the number of unique expected sink compounds recovered in the output. For the catharanthine-vindoline coupling, recovery therefore required two unique sink hits. This scoring confirms recovery of the required precursor compounds but does not necessarily demonstrate that both compounds occurred jointly within the same individual pathway record. Several multi-enzyme transformations involving unstable intermediates were represented as net chemical reactions, including GO-Redox1-Redox2 and T3O-T3R.

2.2 Enzyme characterization

Twenty-nine enzymes associated with the MIA pathway were assessed using protein database records and existing literature. For each enzyme, evidence was collected for direct catalytic activity, recombinant expression, kinetic characterization, substrate specificity, experimental structural information, and mutagenesis or mechanistic studies.

Enzymes were classified as *Well-characterized* when extensive direct experimental evidence was available, including detailed biochemical, structural or mechanistic characterization, and as *Partially characterized* when catalytic activity had been experimentally confirmed but deeper mechanistic, structural or substrate-scope information remained incomplete. The original classification framework additionally allowed inferred and orphan categories, although none of the 29 enzymes in the final dataset fell into these classes. The distinction between direct experimental characterization and prediction by homology was maintained throughout the analysis.

2.3 Protein sequence and homology analysis

Protein sequences were collected for 28 MIA enzymes and combined into a multi-FASTA dataset. PRX1 was retained in the biochemical characterization analysis but excluded from the automated sequence and homology analysis because a confidently assigned full-length protein sequence was not available in the working dataset.

The 28 protein sequences were searched against the NCBI non-redundant protein database using BLASTP through the NCBI web interface, restricted to *Viridiplantae*. The search returned up to 50 hits per query. BLAST output was subsequently processed using custom Python scripts. Hits corresponding to *Catharanthus roseus* were removed, and remaining sequences were considered strong homologues when sequence identity was $\geq 40\%$, query coverage was $\geq 70\%$, and E-value was $\leq 1 \times 10^{-5}$.

Homology summaries were then compared across pathway enzymes to identify relatively conserved and divergent proteins. For the seven most divergent proteins, DAT, SAT, SGD, D4H, T16H, T3R and T3O, the five highest-scoring unique non-*Catharanthus* BLAST hits were selected by bitscore and aligned with the corresponding *C. roseus* query sequence. Mean identity, mean query coverage and fully conserved positions were calculated from these alignments. DAT, the most divergent protein, was then examined in greater detail for the conserved BAHD acyltransferase motifs HXXXD and the C-terminal DFGWG-region motif [10, 11].

2.4 Integrated pathway visualization

RetroPath results and enzyme-characterization data were integrated into a directed pathway network. Metabolites were represented as nodes and biochemical transformations as edges. Edge colour represented RetroPath performance: green for recovered reactions, red for reactions not recovered, and grey for technical errors.

Edge thickness represented enzyme characterization, with thicker edges indicating well-characterized enzymes and thinner edges indicating partially characterized enzymes. For

late-pathway transformations for which no specific enzyme had been resolved, a dashed or dotted edge style was used to indicate unresolved enzyme identity.

Nodes were coloured according to pathway branch, including the iridoid, indole, central, catharanthine, vindoline and bisindole branches. The network was assembled and styled before final graphical editing in Inkscape. The branched topology was retained so that the convergence of secologanin and tryptamine and the later convergence of catharanthine and vindoline remained explicit.

3 Results

3.1 RetroPath performance

A total of 28 chemical transformations were evaluated in the final RetroPath benchmark. Seven reactions produced technical errors and were therefore excluded from the calculation of biological recovery performance. Of the remaining 21 evaluable transformations, 15 were successfully recovered and six were not recovered, corresponding to an overall recovery rate of 71.4%.

Recovered transformations included GES, G8H, 8HGO, TDC, SGD, SAT, ASO, T16H, 16OMT, T3O-T3R, NMT, D4H, DAT, the catharanthine-vindoline coupling reaction, and the transformation of anhydrovinblastine to vinblastine.

In contrast, GS, GO-Redox1-Redox2, DPAS, CS, TS and the vinblastine-to-vincristine transformation were not recovered. ISY, IO, 7DLGT, 7DLH, LAMT, SLS and STR generated technical errors and were therefore treated separately from genuine prediction failures.

Six of the seven reactions that produced technical errors were located in the iridoid branch, suggesting that this region of the pathway was particularly problematic for the local RetroPath implementation. These transformations include chemically complex oxidation, cyclization, glycosylation and methylation steps, which may generate intermediates or reaction-rule matches that are difficult to represent consistently. In several cases, RetroPath failed during rule generation because of chemically invalid structures, including valence errors, while other reactions failed during result construction. These technical errors therefore appear to reflect limitations in reaction-rule generation and molecular representation rather than genuine failure to predict the underlying biochemical transformations.

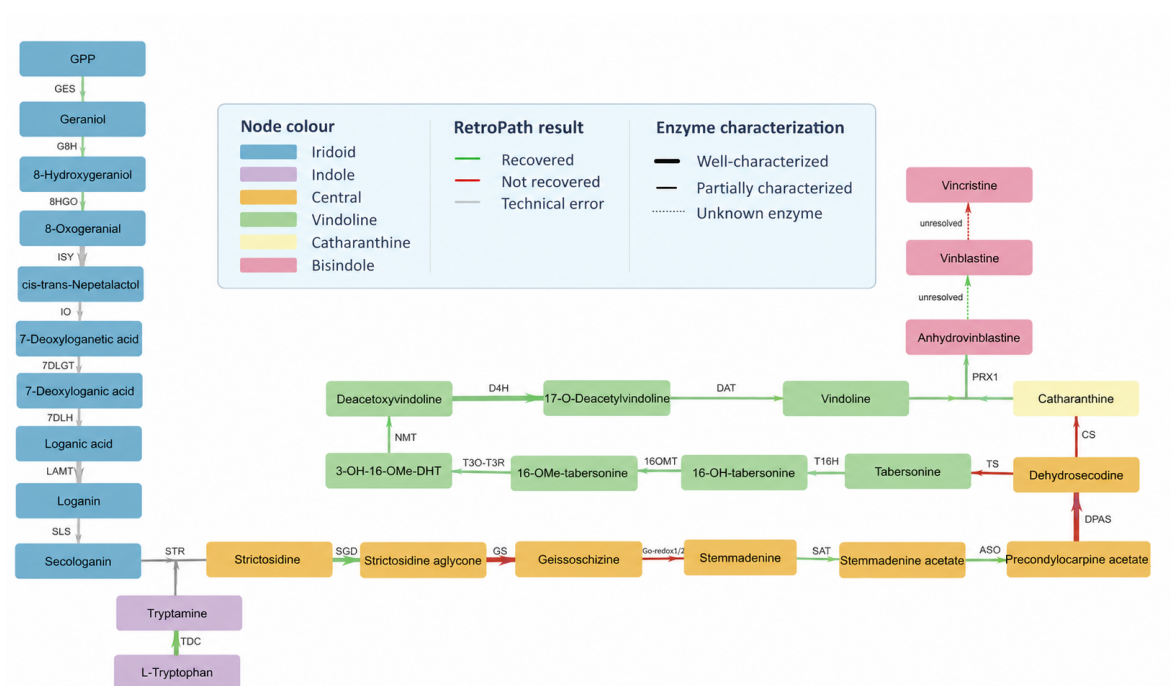


Figure 1: Integrated MIA biosynthetic pathway. Node colours represent pathway branches, edge colours represent RetroPath results, and edge thickness or style represents enzyme characterization status.

3.2 Enzyme characterization

The enzyme-characterization dataset contained 29 enzymes, of which nine were classified as well-characterized and 20 as partially characterized.

PRX1 was retained as a partially characterized enzyme for the coupling of catharanthine and vindoline. By contrast, the enzymes responsible for the subsequent conversion of anhydrovinblastine to vinblastine and vinblastine to vincristine remained unresolved in the present dataset and were therefore not assigned a characterization category.

3.3 Sequence conservation analysis

Protein sequence analysis was performed for 28 enzymes. Seven comparatively divergent proteins, DAT, SAT, SGD, D4H, T16H, T3R and T3O were selected for more detailed comparison.

Among these enzymes, DAT showed the strongest sequence divergence, with a mean identity of 42.1% and mean query coverage of 59.4%. The remaining enzymes showed mean identities between approximately 58.6% and 65.8%, with mean query coverage between 74.6% and 79.9% (Figure 2).

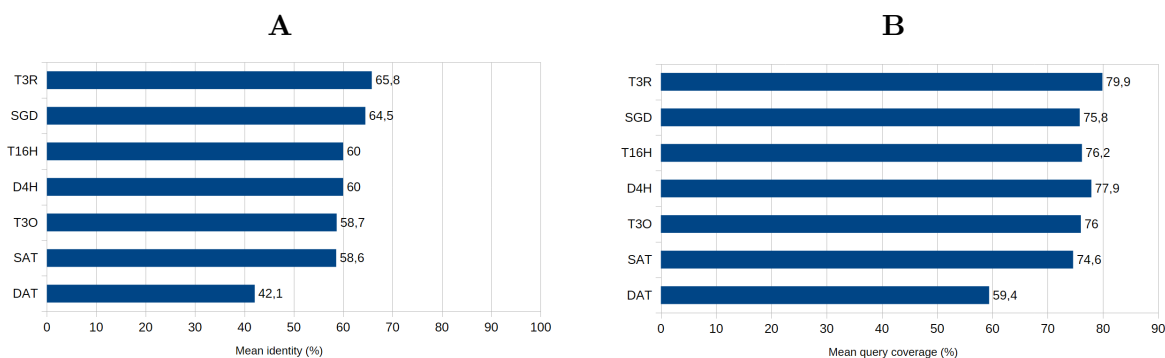


Figure 2: Sequence conservation among selected MIA pathway enzymes. (A) Mean sequence identity of the five selected non-*Catharanthus* homologues for each enzyme. (B) Mean query coverage of the same selected homologues.

The multiple-sequence alignments also showed that DAT contained the lowest number of fully conserved positions, with 109 compared with 140-255 for the other six enzymes. Despite this overall divergence, DAT retained characteristic BAHD acyltransferase sequence motifs, including the catalytic HXXXD motif and the conserved C-terminal DFGWG-region motif (Figure 3).

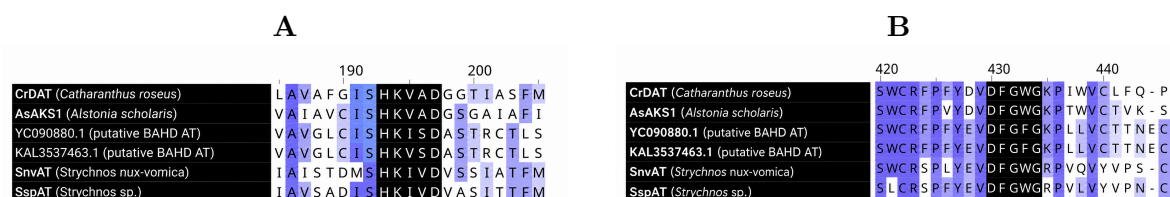


Figure 3: Multiple-sequence alignment of DAT and selected non-*Catharanthus* homologues showing conserved BAHD acyltransferase motifs. (A) Region containing the catalytic HXXXD motif. (B) C-terminal region containing the conserved DFGWG-region motif.

3.4 Integrated pathway analysis

RetroPath performance and enzyme-characterization status were combined into a pathway-level visualization (Figure 1). Edge colour represented computational recovery, while edge thickness or style represented the level of enzyme characterization.

The integrated map showed that several pathway reactions were both computationally recoverable and supported by experimentally characterized enzymes, whereas other steps remained difficult computationally, experimentally, or both.

The late bisindole pathway illustrated this distinction particularly clearly. The PRX1-associated coupling of catharanthine and vindoline was recovered by RetroPath and was associated with a partially characterized enzyme. The subsequent transformation of anhydrovinblastine to vinblastine was also successfully recovered, although the corresponding enzyme remains unresolved. By contrast, the transformation of vinblastine to vincristine was not recovered and its corresponding enzyme also remains unresolved.

4 Discussion

The RetroPath analysis showed substantial variation in performance between different regions of the MIA pathway. Within the vindoline branch, six of seven evaluated transformations were successfully recovered, corresponding to a recovery rate of 85.7%. Transformations catalysed by T16H, 16OMT, T3O-T3R, NMT, D4H and DAT were recovered, whereas the tabersonine synthase-associated transformation was not.

By contrast, six of the seven technical errors observed across the complete benchmark occurred in the iridoid branch. This concentration of technical failures suggests that the local RetroPath implementation encountered particular difficulties when processing this region of the pathway. Iridoid biosynthesis contains transformations involving reductive cyclization, oxidation, glycosylation and other substantial molecular rearrangements. However, because these reactions generated workflow or molecular-representation errors rather than valid RetroPath outputs, the present analysis cannot establish that their underlying chemistry is intrinsically unpredictable [12]. Instead, the results indicate limitations in reaction-rule generation or molecular representation that warrant further investigation.

Taken together, the RetroPath benchmark and enzyme-characterization analyses reveal that computational recoverability and experimental characterization do not always coincide. Several transformations were successfully recovered despite only partial enzyme characterization, whereas other reactions involving experimentally characterized enzymes were not recovered. The technical failures observed particularly in the iridoid branch further demonstrate that inability to evaluate a transformation computationally should not be interpreted as evidence of biochemical infeasibility.

DAT catalyses the final acetylation step in vindoline biosynthesis and was the strongest sequence outlier among the seven proteins examined in greater detail. DAT displayed a mean identity of 42.1% and mean query coverage of 59.4% relative to the selected non-*Catharanthus* homologues. These values indicate comparatively greater sequence divergence than observed for the other six proteins. Importantly, query coverage alone should not be interpreted as evidence for conservation of the protein fold. However, the retention of characteristic BAHD sequence motifs, including HXXXD and the conserved C-terminal DFGWG-region motif, indicates conservation of key sequence features associated with the BAHD acyltransferase family [13].

RetroPath generated 2,861 candidate products for the vinblastine-to-vincristine transformation but failed to recover the expected precursor relationship. Vincristine formation involves modification of the indoline nitrogen, including N-demethylation and formylation. The multistep nature of this chemistry may contribute to its failure to be recovered under the one-step benchmark conditions, although the present analysis cannot determine the precise mechanistic cause of the failure. No corresponding characterized enzyme was identified in the present dataset, making this transformation both a computational and experimental gap in the late MIA pathway.

Importantly, vincristine formation was not the only genuine non-recovery. Six evaluable transformations were not recovered: GS, GO-Redox1-Redox2, DPAS, CS, TS and vincristine formation. These genuine non-recoveries should be distinguished from the seven technical

errors, for which RetroPath did not produce an interpretable prediction.

5 Limitations

RetroPath was run with a maximum pathway length of one reaction, evaluating individual transformations rather than multi-step routes. Longer pathway searches could potentially identify alternative reaction sequences that were not evaluated in the present benchmark.

In addition, BLAST searches were limited to a maximum of 50 returned hits per query. Consequently, homologue counts reflect the sequences returned under the selected search conditions rather than the complete number of homologues present in Viridiplantae.

Seven RetroPath transformations generated technical errors and therefore could not be evaluated for biological prediction accuracy. Future work should identify the specific reaction rules or molecular representations responsible for these failures.

Finally, enzyme-characterization classifications were based on the experimental evidence included in the literature and database records examined during this study and may change as additional biochemical and structural evidence becomes available.

6 Conclusion

This study shows that RetroPath can recover a substantial proportion of individual transformations within the *C. roseus* MIA pathway, with 15 of 21 evaluable transformations successfully recovered. However, performance varied across pathway regions. In particular, the iridoid branch showed a disproportionate concentration of technical failures, indicating difficulties in reaction-rule generation or molecular representation, while several transformations elsewhere in the pathway represented genuine non-recoveries.

Integration of computational performance with enzyme-characterization and sequence data also highlighted unresolved late-pathway steps. The conversion of anhydrovinblastine to vinblastine was computationally recovered despite the absence of a resolved enzyme, whereas vinblastine-to-vincristine formation remained both computationally unrecovered and enzymatically unresolved.

More broadly, systematic benchmarking of retrosynthesis tools against specialized plant metabolism can reveal specific computational and experimental knowledge gaps. Such analyses may help guide both the improvement of reaction-rule databases and the prioritization of enzymes and transformations for future metabolic-engineering studies.

Data Availability

Analysis scripts and processed data are available at: https://github.com/AlexJMilner/MIA_Computational_Assessment

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